

MKID-IFTS Instrument Simulator

Project Overview

High-Resolution Imaging FTS Lab

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1. The Instrument Concept

The High-Resolution IFTS Lab is developing a new class of astronomical instrument that pairs an Imaging Fourier Transform Spectrograph (IFTS) with energy-sensitive Microwave Kinetic Inductance Detectors (MKIDs). This combination addresses a fundamental limitation of existing imaging FTS instruments.

1.1 The Problem with Conventional Imaging FTS

An imaging FTS builds spectra by scanning a Michelson interferometer mirror in discrete steps. The spectral resolution is set by how far the mirror travels (the maximum optical path difference), while the step size determines the Nyquist-sampled bandwidth. When large step sizes are used to achieve high spectral resolution, the sampled bandwidth is narrow and signals from wavelengths outside this band fold on top of each other — a phenomenon called spectral folding or aliasing.

Instruments like SITELE at CFHT solve this with physical bandpass filters: a filter blocks out-of-band light before it enters the interferometer, preventing folding. However, filters restrict each observation to a single narrow wavelength range (typically 30–70 nm), meaning the full visible spectrum (350–900 nm) requires multiple separate observations. Filters also contribute to photon noise through their finite bandpass, and any sky emission within the filter band distributes noise into every spectral channel after Fourier transform.

1.2 The MKID-IFTS Solution

MKIDs are superconducting photon-counting detectors that record not only the arrival time of each photon but also its energy, with a resolving power of $R_e \approx 35\text{--}50$. This intrinsic energy measurement allows each photon to be assigned to the correct folding order computationally, entirely replacing the need for physical bandpass filters.

The result is an instrument that simultaneously covers 350–1100 nm at high spectral resolution, using large mirror step sizes, without any color filters. This represents a fundamental gain: full optical-to-near-IR coverage in a single observation, with the wide contiguous field of view that is the hallmark of imaging FTS instruments.

Because MKIDs are photon-counting detectors, every photon arrives individually with a timestamp and an energy tag. This means order separation is performed per photon before the interferogram is constructed — a fundamentally different approach from CCD-based FTS instruments, where the interferogram is built from integrated exposures and order separation must be done optically with filters.

1.3 Key Design Features

- **Filterless broadband operation:** 350–1100 nm in a single interferogram scan, with order separation performed computationally via MKID energy tags.
- **Dual-output interferometer:** Both complementary output ports feed into the MKID cryostat. Combining the two channels provides $\sqrt{2}$ noise reduction, cancels the DC (zero-frequency) component of the interferogram, and enables correction for atmospheric transparency variations.
- **Offset beam geometry:** The optical design uses an offset geometry rather than a tilted beamsplitter, sending both output beams into the same cryostat while maintaining identical optical path differences at each detector pixel.
- **Optional neutral density filters:** A filter wheel with neutral density filters is available to prevent MKID saturation on very bright sources (MKIDs have a maximum count rate of ~2000–5000 photons/s/pixel), but no wavelength-selective filters are used.

2. The Simulator

2.1 Purpose

Before this instrument is built, its performance must be validated, its configuration optimized, and its science capabilities quantified. The simulator is the tool that accomplishes all three. It models the complete signal chain from astrophysical source to recovered spectrum, incorporating realistic throughput, noise, and the novel MKID-based order-separation process.

2.2 What the Simulator Produces

The simulator functions at three levels of increasing abstraction:

Level 1 — Full simulation: Generates a realistic raw interferogram with Poisson photon noise, applies MKID energy tagging with finite resolution on a per-photon basis, separates orders using one of two data-reduction strategies, performs the Fourier transform with phase correction and optional apodization, and recovers the final broadband spectrum with order stitching. This mode is used for detailed instrument validation and strategy comparison.

Level 2 — Analytical SNR model: Computes the signal-to-noise ratio per spectral channel using closed-form expressions for signal, photon noise, distributed sky noise, and order cross-contamination. This mode executes in under one second and is used for parameter sweeps and optimization studies.

Level 3 — Exposure Time Calculator (ETC): The astronomer-facing tool. Given a source, a target SNR, and a science goal, it returns the required integration time and recommended instrument configuration. It also warns when source brightness exceeds the MKID count-rate ceiling.

2.3 Signal Chain

Stage	Component	Effect on Signal
1	Astrophysical source	Input flux $S(\lambda)$ in photons/s/cm ² /nm
2	Atmosphere	Multiplicative transmission $T_{\text{atm}}(\lambda)$; wavelength-dependent
3	Sky background	Additive: moonlight + OH lines + airglow + thermal
4	Telescope	Collecting area $\times$ mirror reflectivity (per reflection)
5	IFTS optics + modulation	Optics transmission, beamsplitter ME, interferogram generation
6	MKID detection	QE, per-photon energy measurement, dead time, saturation check
7	Order separation	Per-photon assignment to folding orders via energy tags
8	Spectrum recovery	DC removal, phase correction, FFT, apodization, order stitching

9	SNR / ETC	Noise budget per channel; exposure time computation
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3. Research Questions

The simulator exists to answer the following questions, which cannot be resolved analytically and have not been addressed in the literature for an MKID-IFTS instrument.

3.1 Can MKID energy resolution replace physical filters?

At $R_e \approx 35\text{--}50$, adjacent folding orders partially overlap in the MKID's energy measurement. Some photons near order boundaries will be misassigned, introducing cross-contamination noise into the recovered spectrum. The central question is whether this contamination remains small enough that the filterless broadband advantage is preserved. The simulator maps SNR as a function of R_e , identifying the minimum energy resolution required for the order-sorting approach to outperform a filter-based instrument like SITELLE. Note that the achievable R_e depends on photon energy (better in the UV, worse in the IR due to quasiparticle statistics), so this threshold may be wavelength-dependent.

3.2 Which data reduction strategy produces less noise?

Two strategies exist for handling photons near order boundaries:

- **Strategy 1 (hard-cut filtering):** Photons whose MKID-measured energy falls within a grey zone near an order boundary are discarded. This eliminates contamination but sacrifices signal. The grey-zone width (in units of σ_E) is a tunable parameter.
- **Strategy 2 (probabilistic weighting):** Each photon contributes to multiple orders weighted by the probability from its energy measurement kernel. All photons are retained, but correlated noise is introduced between adjacent orders.

The simulator implements both strategies and compares their output SNR across a range of source types (emission-line nebulae, faint continuum galaxies, cosmological targets) and MKID energy resolutions. This comparison is a primary scientific output of the project. The answer may depend on source type: bright emission-line sources may favor the probabilistic approach (maximizing photon retention), while faint continuum sources near the sky background may favor hard cuts (minimizing contamination from sky-dominated adjacent orders).

3.3 What is the optimal instrument configuration?

The folding order (which sets the step size and the number of orders spanning the full bandwidth), the number of mirror steps (which sets the spectral resolution within each order), and the ZPD scan fraction are all free parameters. The optimal combination depends on the science case. More folding orders means larger step sizes and higher spectral resolution per order for the same number of steps, but also more order boundaries and therefore more cross-contamination. The simulator maps this configuration space to identify the optimum for each science scenario.

3.4 How does the instrument perform at the red edge?

The 700–1100 nm region is dominated by hundreds of bright OH airglow lines. In a conventional FTS, the photon noise from these lines distributes into every spectral channel (the distributed noise property). MKID order-sorting confines each OH line’s noise contribution to its own order, potentially providing a significant advantage. The simulator quantifies this by comparing the SNR of a faint source at 900 nm with and without MKID order-sorting, against the full OH sky background.

3.5 How robust is performance to mirror positioning errors?

OPD jitter from imperfect mirror positioning introduces phase errors into the interferogram, broadening the instrument line shape and reducing SNR. The simulator models this by adding random Gaussian perturbations to the mirror positions, generating multiple realizations, and measuring the degradation as a function of jitter amplitude. Results will be compared against the SITELLE benchmark of ~20–30 nm RMS producing < 1% degradation.

4. Known Simplifications

The following effects are acknowledged but intentionally deferred from the initial simulator version. They do not change the core science conclusions but would add significant development complexity:

- **Time-variable atmosphere:** Atmospheric transmission is treated as static during an observation. Real variability (clouds, changing airmass) is correctable via the dual-output port sum.
- **2D spatial effects:** The simulator operates per-pixel (1D). The off-axis geometry causes a mild variation in effective OPD and spectral resolution across the field; this can be captured as a field-position scaling factor applied to the 1D result.
- **Cosmic ray glitches:** Cosmic ray hits produce large quasiparticle bursts in MKIDs, temporarily blinding individual pixels. In real data these are flagged and the affected interferogram steps are masked. The simulator does not model these transients but the noise budget should note them as an unmodeled systematic.
- **Wavelength calibration errors:** The simulator assumes perfect knowledge of mirror step positions. Real instruments use laser calibration cubes to determine the OPD-to-wavenumber mapping. A calibration error placeholder exists in the architecture for future implementation.
- **Flat-field variation:** Throughput varies across the field of view (maximum at center, lower in corners). The simulator computes the on-axis case; a flat-field map can be applied as a divisive correction to estimate corner performance.

5. Science Impact and Deliverables

5.1 For the instrument team

- Quantitative validation that the MKID-IFTS concept works before committing to hardware.
- Optimized instrument parameters (folding order, step size, beamsplitter specification) for the target science cases.
- Recommended data reduction strategy with quantified performance gains.
- Error budget: contribution of each noise source to the total noise, including order contamination.
- Saturation and dynamic range limits: maximum source brightness per pixel before count-rate nonlinearity degrades performance.

5.2 For the astronomical community

- An Exposure Time Calculator that allows any astronomer to assess observability and plan integration time, with a simple interface requiring only source type, magnitude, and science goal.
- Quantitative comparison against SITELE for standard science cases.
- Published open-source simulator code as the reference performance model.

5.3 Deliverables

1. Installable Python package implementing the full end-to-end simulator.
2. Exposure Time Calculator with astronomer-facing interface (command-line minimum; optional web or notebook GUI).
3. Technical report: Strategy 1 vs. Strategy 2 comparison across source types and R_e values.
4. Configuration optimization maps for at least three science cases.
5. Demonstration Jupyter notebooks with worked examples.